BSc (HONS) in Computing 2nd - Year

Data Structures & Algorithm

CA – 1

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Project Proposal

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Project Name: Smart Docklands Food Truck Tracker

Theme: Smart Economy/ Smart Community

This initiative concentrates on the improvement of urban services and support for innovative projects within the Docklands area. One main important aspect of this is supporting. One important aspect of this is supporting the local businesses and enhancing the lives of the residents, workers and visitors. Food trucks are gaining more and more popularity in urban areas because they offer fast and varied food choices. However, it can often be difficult for people to know which food trucks are available, what type of food they sell, and where they are located at a given time.

The purpose of this project is to design and implement a Java application known as **Smart Docklands Food Truck Tracker.** The purpose of this application is to manage food trucks operating in the Docklands area and allow users to view and interact with information about them. The system will allow food trucks to be registered in the application and stored in an internal data structure. Users will be able to view the available trucks, search for a specific truck, and remove trucks when necessary.

The system will be developed using Java and will include a **Swing graphical user interface (GUI)** that allows users to interact with the system through buttons and text fields. The GUI will allow users to input information such as the name of a food truck, its cuisine type, and its location. This information will then be stored as objects within the program.

The application will make use of the use of **Abstract Data Types (ADTs)** to manage the data in an efficient way. A **Singly Linked List** will be used to store the list of food trucks in the system. Each node in the list will store an object for a food truck and the reference for the next node in the list.

A **Stack data structure** will also be implemented to keep track of recently added food trucks. This will be a stack following **the Last-In-First-Out (LIFO)** principle; i.e. the most very latest truck added will be the first one to be removed in case of necessity. In addition, a **Queue data structure** will be used to store customer orders. The queue will behave on **the First-In-First-Out (FIFO)** principle so that the orders are processed in the same order they were received.

The system will demonstrate the use of **interfaces** to define the behaviours of the ADTs used in the project. Interfaces like a **stack interface** and **list interface** will define the methods that are needed to manipulate the data structures. The project will also give an example of inheritance and polymorphism using a base class representing a vehicle in general and a subclass representing a food truck.

The application will support the **CRUD operations** (Create, Read, Update, Delete) of food truck objects. Users will be able to add new food trucks, view the list of trucks, update information, and remove trucks from the system. These operations will be accessed through the graphical user interface.

Overall, this is a good project that demonstrates how data structures and algorithms can be applied in a real-life scenario. By using linked lists, stacks, queues, and object-oriented

programming concepts, the Smart Docklands Food Truck Tracker provides a simple system for managing food trucks and improving access to information within the Docklands area.
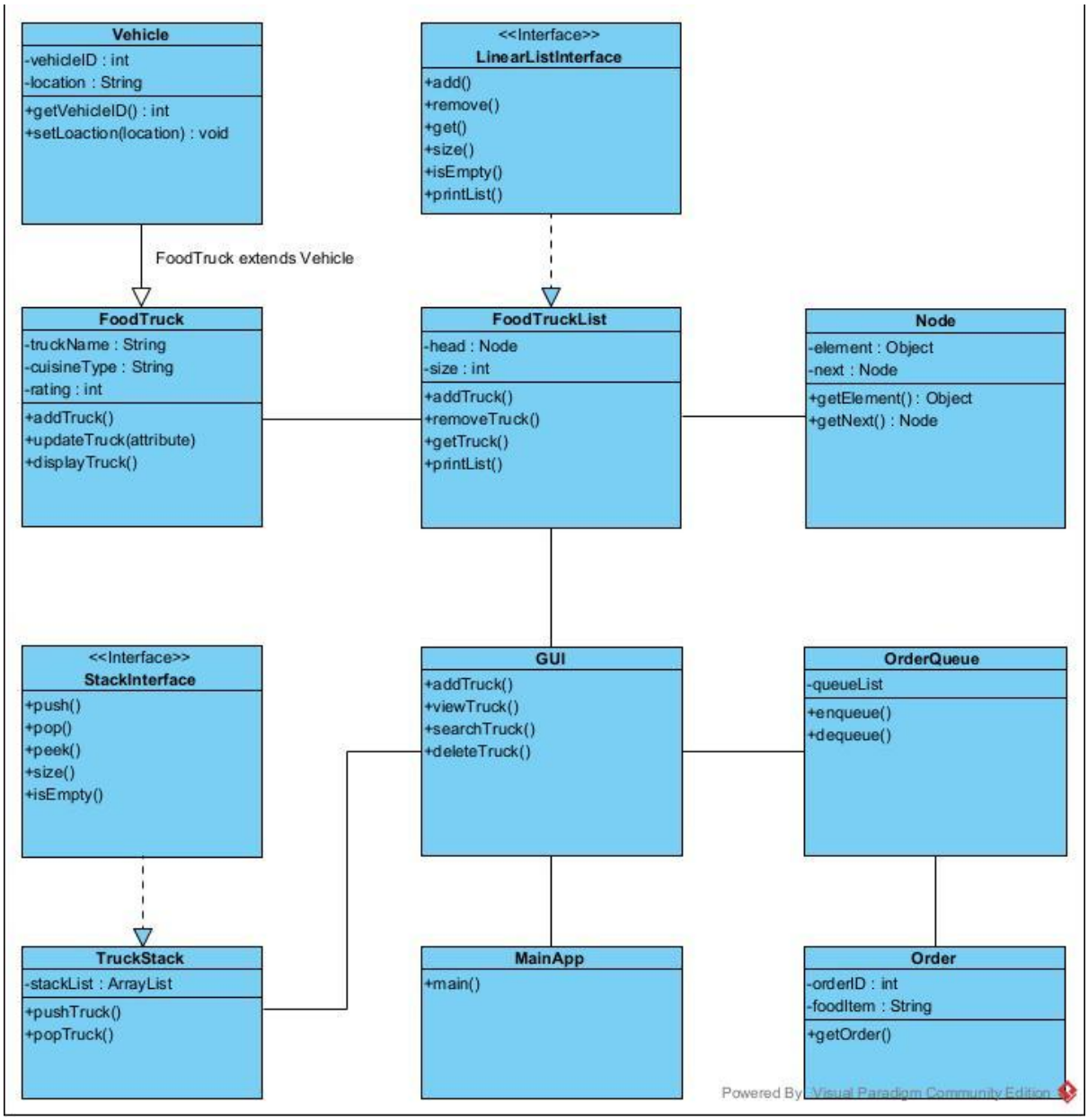
Figure 1: UML Class Diagram for the Smart Docklands Food Truck Tracker System :